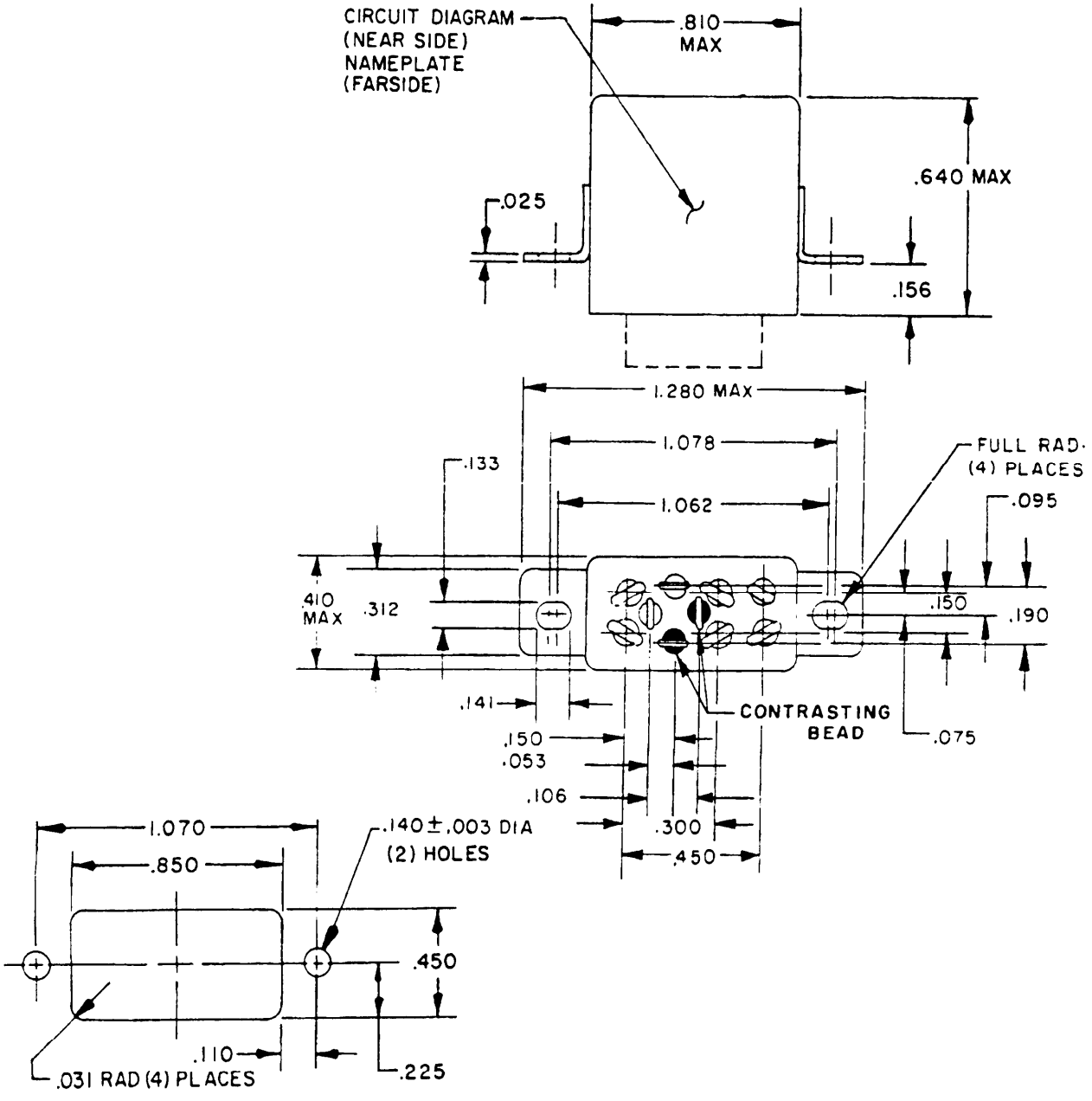
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC (TYPE I), MAGNETIC LATCH (WITH INTERNAL DIODES FOR TRANSIENT SUPPRESSION), PERMANENT MAGNET DRIVE, LOW LEVEL TO 5 AMPERES, 2 PDT, ALL WELDED, HERMETICALLY SEALED

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-6105.



MOUNTING HOLE LAYOUT

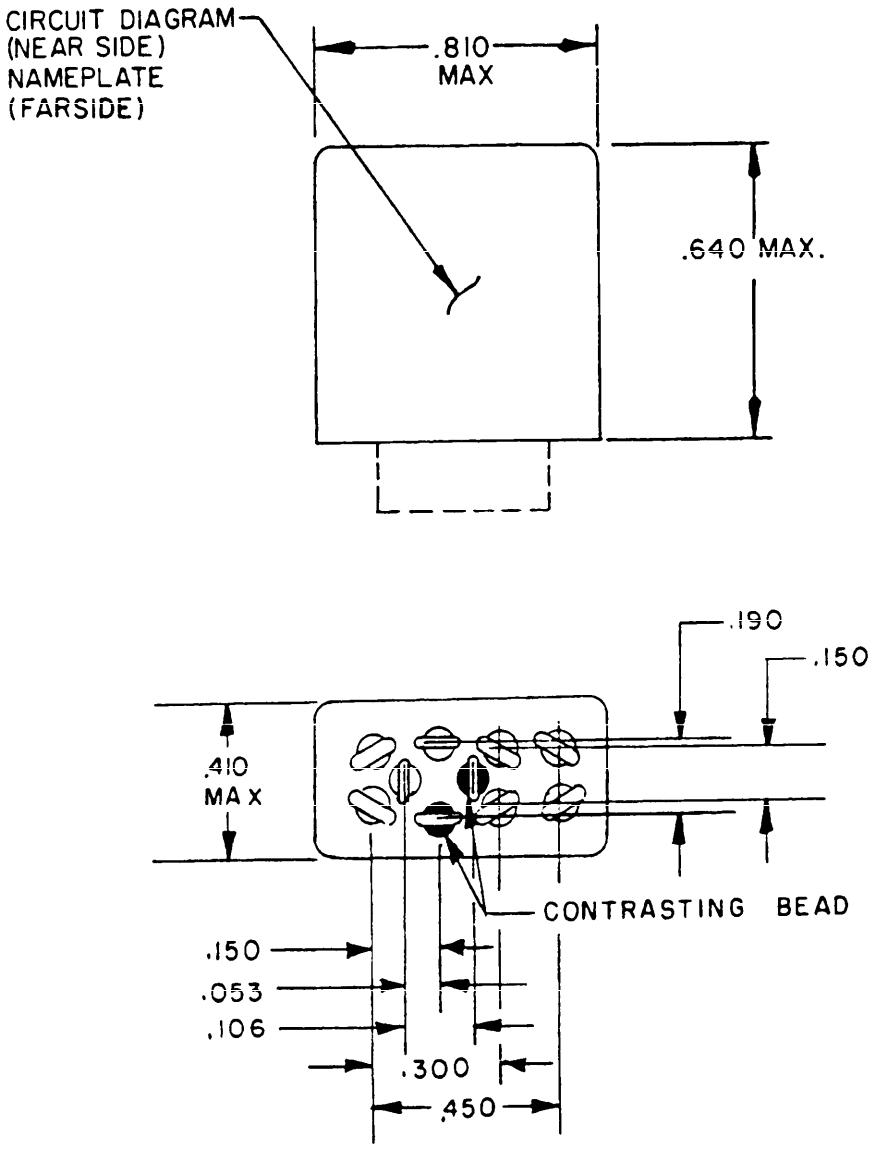
CONFIGURATION A

FIGURE 1. Outline drawing.

AMSC N/A FSC 5945

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

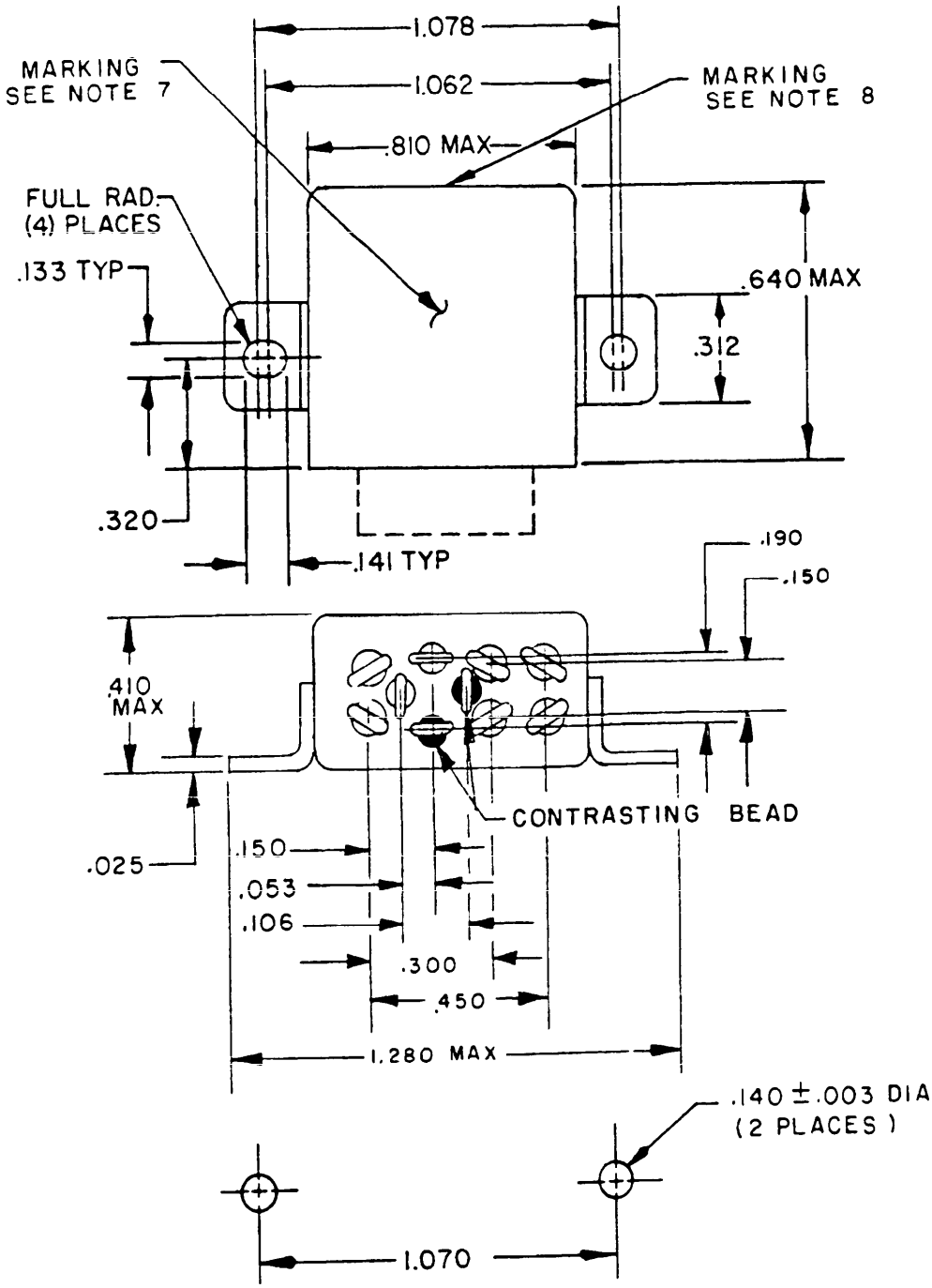
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CONFIGURATION B

FIGURE 1. Outline drawing - Continued.

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MOUNTING HOLE LAYOUT

CONFIGURATION C

FIGURE 1. Outline drawing - Continued.

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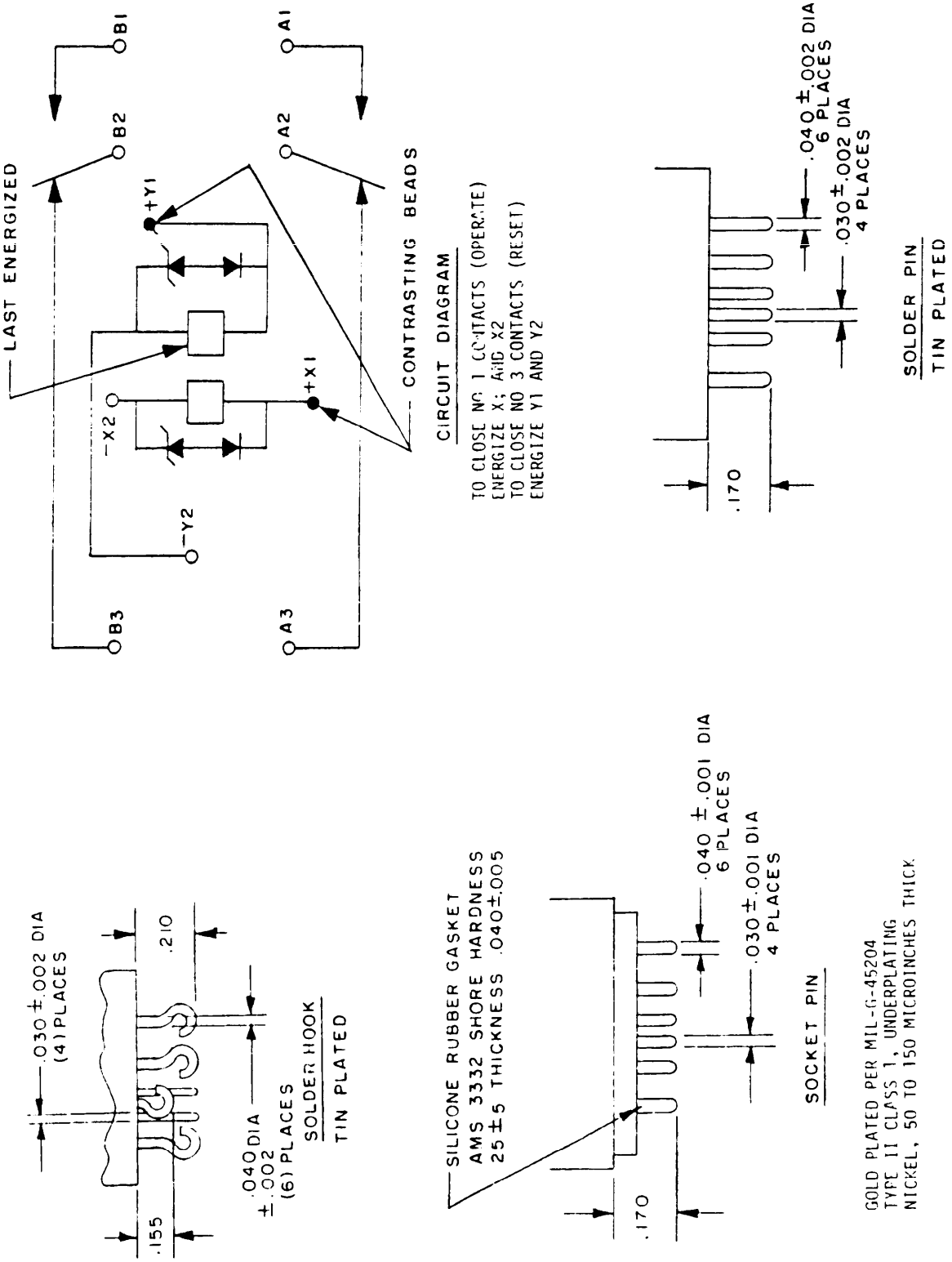


FIGURE 1. Outline drawing - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.106	2.69	.300	7.62
.002	0.05	.133	3.38	.312	7.92
.003	0.08	.140	3.56	.410	10.41
.005	0.13	.141	3.58	.450	11.43
.025	0.64	.150	3.81	.640	16.26
.030	0.76	.155	3.93	.810	20.57
.031	0.79	.156	3.96	.850	21.59
.040	1.02	.170	4.32	1.062	26.97
.053	1.34	.190	4.82	1.070	27.18
.075	1.90	.210	5.33	1.078	27.38
.095	2.52	.255	5.72	1.280	32.51

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. These relays are polarized bistable.
6. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electro-magnet flux.
7. The circuit diagram, manufacturer's part number, and the data shall be marked on the far side.
8. Relays shall be marked with the manufacturer's name or source code and data code. Marking shall be with the bottom of the print adjacent to the near side.

FIGURE 1. Outline drawing - Continued.

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REQUIREMENTS:

Coil data:

Coil data: See table I.

Duty rating: Continuous.

Operational data:

Rated contact load (endurance): See table II.

Operate time: See table I.

Reset time: See table I.

Circuit breaker compatibility: Applicable. 1/

Physical:

Dimensions and configuration: See figure 1.

Weight (maximum): .036 pound (16.34 grams).

Terminal type: See figure 1 and table I.

Finish: Paint optional.

ENVIRONMENTAL CHARACTERISTICS:

Temperature rating: -70°C to +125°C.

Maximum altitude: 300,000 feet.

Shock g-level: 200 g's.

Duration: 6 milliseconds.

Maximum duration contact opening: 10 µs.

Vibration (sinusoidal):

G-level: 30 g's.

Frequency range: 70 - 3000 Hz.

Vibration (random):

Applicable standard: MIL-STD-202.

Method 214: 0.4 g^2/Hz.

Test condition: IG.

Duration: 15 minutes each plane.

Acceleration: 15 g's.

1/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence as listed in table III. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

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ELECTRICAL CHARACTERISTICS:

Insulation resistance:

Initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level):

	Initial	After life tests
Coil to case	500 V rms	350 V rms
Coil to coil	500 V rms	350 V rms
All other points	1000 V rms	750 V rms

Dielectric withstanding voltage (altitude) 2/:

	80,000 ft	300,000 ft
Coil to case	200 V rms	375 V rms
Coil to coil	200 V rms	375 V rms
All other points	250 V rms	500 V rms

Maximum contact drop:

Initial: 0.150 volt.

After life test: 0.175 volt.

Overload current: 20 amperes dc, 30 amperes ac.

Rupture current: 25 amperes dc, 40 amperes ac.

Part number: M6106/54-(dash number from table I).

2/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

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TABLE I. Operating characteristics.

Dash number	Coil data: Coil 2/	Nominal: Volts	Nominal: Freq. Hz	Nominal: Res. ±10% at 25°C 3/	Max: Volts	Max: Amperes at 25°C	Max pick-up voltage: Normal 1/	Max pick-up voltage: High temp test	Max pick-up voltage: Cont current test	Time (milliseconds, max): Operate	Time (milliseconds, max): Reset	Contact bounce: Main NO	Contact bounce: Main NC	Contact bounce: Aux NO	Contact bounce: Aux NC	Terminal type	Mating socket	Configuration
-001	X1 X2 Y1 Y2	28	DC	730	29	.044	18	19.8	22.5	4	4	.5	.5	--	--	Solder hook	N/A	A
-002 4/																		
-003	X1 X2 Y1 Y2	28	DC	730	29	.044	18	19.8	22.5	4	4	.5	.5	--	--	Socket pin	MIL-S-12883/45-004	A
-004 4/																		
-005	X1 X2 Y1 Y2	28	DC	730	29	.044	18	19.8	22.5	4	4	.5	.5	--	--	Solder pin	N/A	A
-006 4/																		
-007	X1 X2 Y1 Y2	28	DC	730	29	.044	18	19.8	22.5	4	4	.5	.5	--	--	Solder hook	N/A	C
-008 4/																		
-009	X1 X2 Y1 Y2	28	DC	730	29	.044	18	19.8	22.5	4	4	.5	.5	--	--	Solder pin	N/A	B
-010 4/																		
-011	X1 X2 Y1 Y2	28	DC	730	29	.044	18	19.8	22.5	4	4	.5	.5	--	--	Solder pin	N/A	C
-012 4/																		

1/ Over the temperature range.
2/ Back EMF 42 V dc maximum.
3/ Coil resistance not directly measureable at coil terminals.
4/ Screened relays shall be tested to the group A, ER requirements only and additional life testing is not required. For group A inspection, operational reliability shall be performed at the low level run-in.

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TABLE II. Rated contact load endurance (amperes per pole) case grounded.

Type of load	Endurance (life) operating cycles x 10^3	28 V dc		115 V ac, 1 phase	115/200 V ac, 3 phase	See appropriate notes
		Main		Main	Main	
		NO	NC	400 Hz	400 Hz	
Resistive	100	5	5	5	5	
Inductive	100	-	-	-	-	
Inductive	20	3	3	5	5	
Motor	100	2	2	3	3	
Lamp	100	1	1	1	1	
Transfer load	-	-	-	-	-	1/
Mechanical life reduced current	400	1.25	1.25	1.25	1.25	
Intermediate current	50	Applicable per specification				
Low level	100	Applicable per specification				2/, 3/
Multiple loads	As applicable					4/

1/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

2/ For low level, the following shall apply:
   a. During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes; 10- to 50-millivolt open circuit voltage, 100-ohm maximum contact resistance.
   b. Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum.

3/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.

4/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts.

TABLE III. Current level (resistive) and duration.

1	10 amperes	1 hour
2	15 amperes	20 seconds
3	37.5 amperes	2 seconds
4	75 amperes	.53 second

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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